

Knowledge Representation

Lecture 12: Default Logics

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Traditional vs. human reasoning

- Traditional logics
 - deal with *complete* information
 - are *monotonic*: adding new premises never invalidate old conclusions
 - the underlying concept is that of *truth*.

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 - the underlying concept is that of *truth*.
- Human reasoning
 - deals with *incomplete* information
 - conclusions need not be correct and may be *invalidated* in view of new evidences
 - is grounded on the concept of *rationality*.

Rationality

Rationality is:

- **agent-oriented**: different agents have usually different opinions as to what is (or is not) rational in a given situation
- **purpose-oriented**. E.g. one may assume that *Bill* is honest and lend him 50\$, but would be more cautious while considering him as a potential business partner.

Common-sense conclusions are called **beliefs**.

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- **purpose-oriented**. E.g. one may assume that *Bill* is honest and lend him 50\$, but would be more cautious while considering him as a potential business partner.

Common-sense conclusions are called **beliefs**.

Definition 9.1. (Perlis) A statement α is believed by an agent Ag , i.e. he views it as his belief, iff he is prepared to use α (act, take decisions) as if it were true.

Example 12.1 (Winograd, 1980)

Bill is planning to make a trip by car. Given no other evidence, he assumes that

α : *his car is where it was parked.*

- If *Bill* ignores all circumstances under which α could be false and base his actions on the assumption that it is true, then α is his belief.
- However, even if he is almost sure that α is true, but anyway tries to improve his chances and check whether a bus will pass by soon, then he does not regard α as his belief, but rather as a very likely contingency.

Non-monotonicity

- Traditional deduction is **monotonic** – new premises cannot invalidate old conclusion. Formally,

$$T_1 \subseteq T_2 \implies Th(T_1) \subseteq Th(T_2),$$

where $Th(T) = \{\alpha : T \vdash \alpha\}$.

Common-sense reasoning is **non-monotonic** – new premises may invalidate previously derived conclusions.

- Non-monotonic rule:

Given α , in the absence of evidence β , infer γ .

α supports the conclusion γ , while the lack of β assures its rationality.

Non-monotonicity and RAC

- *Frame problem* – all aspects of the world remain invariant except from those explicitly changed by actions. In other words, in the absence of evidence on the contrary, accept only those changes that are caused by performing actions.
- *Qualification problem* – in the absence of evidence on the contrary, assume that the action succeeds and its execution leads to given effects.

Typology of non-monotonicity

(1) Default reasoning: e.g. *In the absence of evidence on the contrary, assume that a bird can fly.*

(2) Autoepistemic reasoning: e.g. *Unless your name is on a list of winners, assume that you are a loser.*

Typology of non-monotonicity

(1) Default reasoning: e.g. *In the absence of evidence on the contrary, assume that a bird can fly.*

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(1) deals with incomplete information, while **(2)** refers to complete information (list of winners), but incomplete representation (lack of the list of losers).

Types of default reasoning

Types of reasoning	Explanation of the conclusion γ
<i>Prototypical reasoning</i>	γ describes a typical situation, supported by statistical observations, so there are good chances that γ holds.
<i>No-risk reasoning</i>	If $\neg\gamma$ leads to dangerous consequences, then γ should be accepted.
<i>Best-guess reasoning</i>	In view of the available information, γ is the best guess to make.
<i>Probabilistic default reasoning</i>	Assuming that probabilistic values are sufficiently high, it is sensible to infer γ .

Definition 9.2. (Reiter, 1980) A *default* is an expression of the form

$$\delta(x) = \frac{\alpha(x) : \beta_1(x), \dots, \beta_n(x)}{\gamma(x)}$$

where $\alpha(x), \beta_1(x), \dots, \beta_n(x), \gamma(x)$ are first-order formulas with free variables $x = (x_1, \dots, x_k)$. We say that

- $\alpha(x)$ is a *prerequisite*
- $\beta_1(x), \dots, \beta_n(x)$ are *justifications*, and
- $\gamma(x)$ is a *consequent* of the default $\delta(x)$.

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Intuitively,

For all individuals $x = (x_1, \dots, x_k)$, if $\alpha(x)$ is believed and all $\beta_1(x), \dots, \beta_n(x)$ are consistent with what is believed, then $\gamma(x)$ is to be believed.

Open and close defaults

A default $\delta(x)$ is **open** iff it contains at least one free variable; otherwise it is called **closed**.

For example,

$$\frac{Bird(x) : CanFly(x)}{CanFly(x)}$$

open default

$$\frac{Bird(tweety) : CanFly(tweety)}{CanFly(tweety)}$$

closed default.

Open defaults represent universally quantified default rules. By substituting variables by well-founded terms (terms without variables) we obtain closed defaults being **instances** of open ones.

Examples (cont.)

- *A typical bird can fly:*

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- *Adults are usually married unless they are students:*

$$\frac{Adult(x) : Married(x) \wedge \neg Student(x)}{Married(x)}$$

Examples (cont.)

- *A typical bird can fly:*

$$\frac{Bird(x) : CanFly(x)}{CanFly(x)}$$

- *Adults are usually married unless they are students:*

$$\frac{Adult(x) : Married(x) \wedge \neg Student(x)}{Married(x)}$$

- *Typically, children have parents:*

$$\frac{: Child(x) \rightarrow hasParents(x)}{Child(x) \rightarrow HasParents(x)}$$

Applicability of defaults

Syntactically, a closed default

$$\delta = \frac{\alpha : \beta_1, \dots, \beta_n}{\gamma}$$

is applicable wrt the set S of closed formulas (beliefs) iff:

- $\alpha \in S$
- for every $i = 1, \dots, n$, $\neg\beta_i \notin S$.

If δ is applicable wrt S , then $\gamma \in S$.

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Semantically, δ is applicable wrt the set M of models iff

- each model $m \in M$ is a model of α
- for every $i = 1, \dots, n$, β_i has a model in M .

Default Theory

Definition 9.3. By a *default theory* we mean a pair $T = (A, \Delta)$, where A is the set of closed first-order formulas and Δ is the set of defaults.

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A *closure* of an open default theory $T = (A, \Delta)$ is obtained by substituting all open defaults from Δ by their instances wrt all grounded terms (terms without variables).

Example 12.2: Italians

Consider the following statements:

- *Marco, Carlo, and Giuseppe are Italians.*
- *Typically, Italians are Catholics.*

Representation in DL:

$$A = \{Italian(m) \wedge Italian(c) \wedge Italian(g)\}$$

$$\Delta = \left\{ \frac{Italian(x) : Catholic(x)}{Catholicx} \right\}.$$

From $T = (A, \Delta)$ we can infer that all three Italians are Catholics.

Example 12.2 (cont.)

Assume that we found out that *Giuseppe is not a Catholic*.

Representation in DL:

$$A = \left\{ \begin{array}{l} Italian(m) \wedge Italian(c) \wedge Italian(g) \\ \neg Catholic(g) \end{array} \right\}$$
$$\Delta = \left\{ \frac{Italian(x) : Catholic(x)}{Catholic(x)} \right\}.$$

Now we have to reject our previous conclusion that Giuseppe is a Catholic.

Example 12.2 (cont.)

Assume now that we learn that *Marco is a communist*.

Representation in DL:

$$A = \left\{ \begin{array}{l} Italian(m) \wedge Italian(c) \wedge Italian(g) \\ \neg Catholic(g) \\ Communist(m) \end{array} \right\}$$

$$\Delta = \left\{ \frac{Italian(x) : Catholic(x)}{Catholic(x)} \right\}.$$

We can still derive that *Marco is a Catholic*.

Example 12.2 (cont.)

This problematic conclusion can be blocked in various ways.

(1) *All communists are not Catholics.*

Representation in DL:

$$A_1 = \left\{ \begin{array}{l} Italian(m) \wedge Italian(c) \wedge Italian(g) \\ \neg Catholic(g) \\ Communist(m) \\ \forall x. Communist(x) \rightarrow \neg Catholic(x) \end{array} \right\}$$
$$\Delta = \left\{ \frac{Italian(x) : Catholic(x)}{Catholic(x)} \right\}.$$

This way we can conclude that *Marco is not a Catholic.*

Example 12.2 (cont.)

(2) Typically, an Italian is a Catholic unless he is a communist.

Representation in DL:

$$A = \left\{ \begin{array}{l} Italian(x) \wedge Italian(c) \wedge Italian(g) \\ \neg Catholic(g) \\ Communist(m) \end{array} \right\}$$
$$\Delta' = \left\{ \frac{Italian(x) : Catholic(x) \wedge \neg Communist(x)}{Catholic(x)} \right\}.$$

Now we can infer neither $Catholic(m)$ nor $\neg Catholic(m)$.

Example 12.2 (cont.)

(3) Typically, communists are not Catholics.

$$A = \left\{ \begin{array}{l} Italian(m) \wedge Italian(c) \wedge Italian(g) \\ \neg Catholic(g) \wedge Communist(m) \end{array} \right\}$$

$$\Delta'' = \left\{ \frac{Italian(x) : Catholic(x)}{Catholic(x)}, \frac{Communist(x) : \neg Catholic(x)}{\neg Catholic(x)} \right\}.$$

Example 12.2 (cont.)

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Now there is a conflict:

- From the first default we get $Catholic(m)$ (as an Italian)
- From the second default we get $\neg Catholic(m)$ (as a communist).

Both defaults cannot be applied simultaneously, but separately. Then there are two sets of beliefs, B_1 and B_2 , such that $Catholic(m) \in B_1$ and $\neg Catholic(m) \in B_2$.

Extensions

Given a default theory $T = (A, \Delta)$, the set of all formulas obtained either by classical deduction or by applying defaults from Δ is called an *extension of T* .

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Given $T = (A, \Delta)$, the natural requirements imposed on the extension $E(T)$ of T are the following *closure conditions*:

- $A \subseteq E(T)$,
- $E(T) = ThE(T)$,
- $E(T)$ should be the maximal set of formulas containing conclusions of all applicable defaults.

Example 12.3: Tweety Example

Consider the sentences:

- *Tweety is a bird.*
- *A typical bird can fly.*

Representation in DL:

$$A = \{Bird(tweety)\}, \Delta = \left\{ \frac{Bird(tweety) : CanFly(tweety)}{CanFly(tweety)} \right\}.$$

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Representation in DL:

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There are two sets satisfying closure conditions:

- $E_1 = Th(\{Bird(tweety), CanFly(tweety)\})$
- $E_2 = Th(\{Bird(tweety), \neg CanFly(tweety)\})$,

but only the first one is intuitively justified as a set of beliefs.

Extension (cont.)

Definition 9.4. Let $T = (A, \Delta)$ be a closed default theory and let S be a set of closed formulas. By $\Gamma(S)$ denote the smallest set of formulas satisfying:

- $A \subseteq \Gamma(S)$
- $\Gamma(S) = Th(\Gamma(S))$
- if
 - $\frac{\alpha : \beta_1, \dots, \beta_n}{\gamma} \in \Delta$
 - $\alpha \in \Gamma(S)$
 - $\neg\beta_1, \dots, \neg\beta_n \notin S$then $\gamma \in \Gamma(S)$.

The set E of formulas is an **extension of T** iff $E = \Gamma(E)$, i.e. it is a fixed point of the operator Γ .

Example 12.3 (cont.)

$$A = \{Bird(t)\}, \quad \Delta = \left\{ \frac{Bird(t) : CanFly(t)}{CanFly(t)} \right\}.$$

Consider $E_1 = Th(\{Bird(t), CanFly(t)\})$.

- $\Gamma(E_1) = Th(\{\dots\})$, since $E = Th(E)$
- $\Gamma(E_1) = Th(\{Bird(t), \dots\})$, since $A \subseteq \Gamma(E)$
- $\frac{Bird(t) : CanFly(t)}{CanFly(t)} \in \Delta, Bird(t) \in \Gamma(E_1), \neg CanFly(t) \notin E_1 \implies CanFly(t) \in \Gamma(E_1)$

Since $\Gamma(E_1) = Th(\{Bird(t), CanFly(t)\}) = E_1$, this is indeed the extension of (A, Δ) .

Example 12.3 (cont.)

$$A = \{Bird(t)\}, \quad \Delta = \left\{ \frac{Bird(t) : CanFly(t)}{CanFly(t)} \right\}.$$

Consider $E_2 = Th(\{Bird(t), \neg CanFly(t)\})$.

- $\Gamma(E_2) = Th(\{Bird(t), \dots\})$
- $\neg CanFly(t) \in E_2$
- Since $\Gamma(E_2)$ is the SMALLEST set of formulas, $CanFly(t) \notin \Gamma(E_2)$.

$\Gamma(E_2) = Th(\{Bird(t)\}) \neq E_2$, so E_2 is not the extension of (A, Δ) .

No other candidates for extensions exist, so E_1 is the only extension of (A, Δ) .

Example 12.4: Nixon Diamond Problem

Consider the following sentences:

- *Typically, republicans are not pacifists.*
- *Typically, quakers are pacifists.*
- *Nixon is a quaker an a republican.*

Denote:

$R(x)$ – x is a republican

$Q(x)$ – x is a quaker

$P(x)$ – x is a pacifist.

Nixon Diamond Problem (cont.)

Representation in DL:

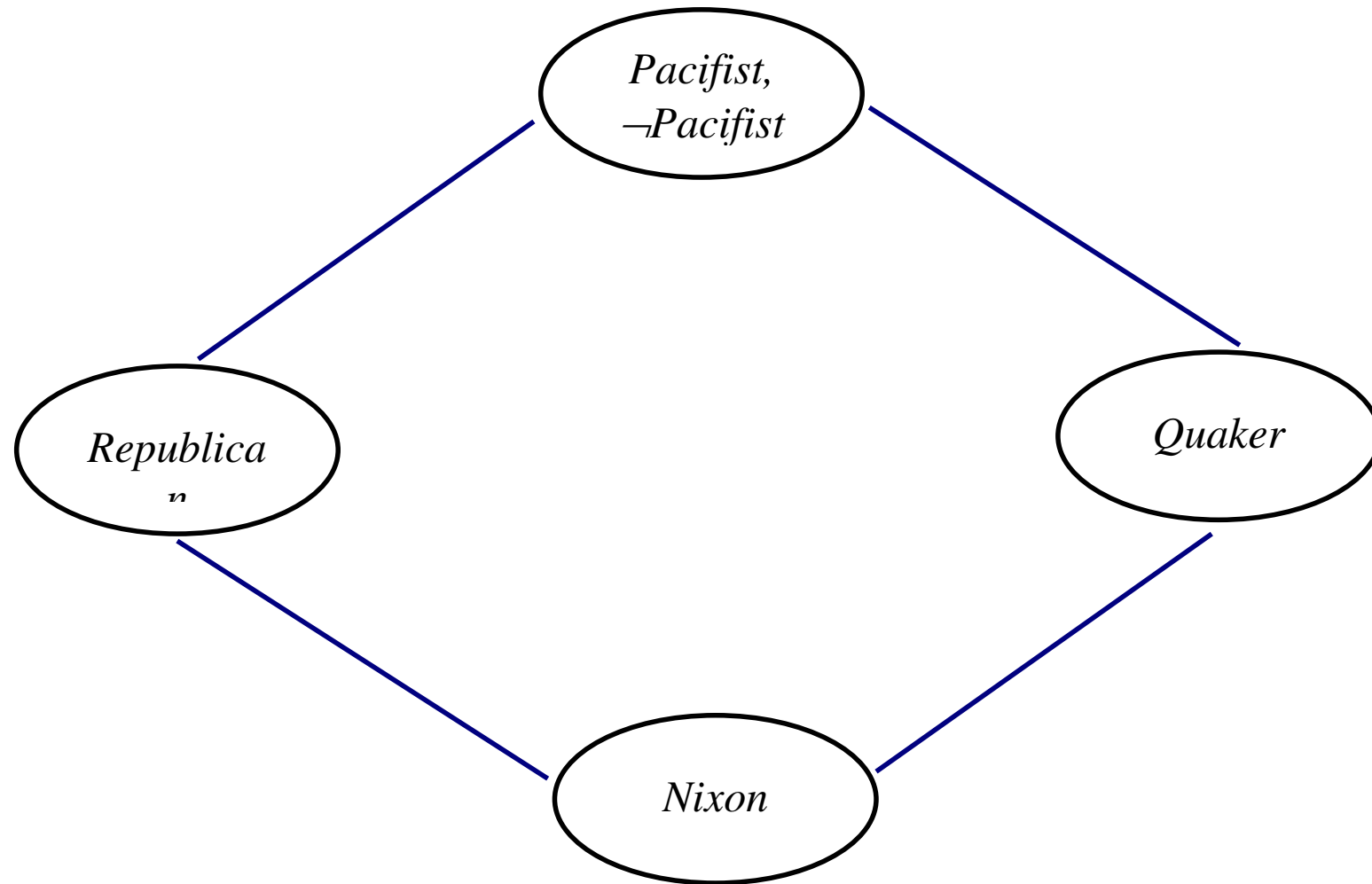
$$A = \{R(n) \wedge Q(n)\}$$

$$\Delta = \left\{ \frac{R(x) : \neg P(x)}{\neg P(x)}, \frac{Q(x) : P(x)}{P(x)} \right\}.$$

We have two extensions of (A, Δ) :

- $E_1 = Th(\{R(n), Q(n), P(n)\})$
- $E_2 = Th(\{R(n), Q(n), \neg P(n)\})$.

Nixon Diamond Problem (cont.)



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- $\delta \in \Delta, \top \in \Gamma(E_1), \neg p \in E_1 \implies \neg p \notin \Gamma(E_1)$.

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- $\delta \in \Delta, \top \in \Gamma(E_1), \neg p \in E_1 \implies \neg p \notin \Gamma(E_1)$.

Hence $\Gamma(E_1) = Th(\emptyset) \neq E_1$, so E_1 is not an extension of (A, Δ) .

Example 12.5 (cont.)

$$A = \emptyset, \quad \Delta = \left\{ \delta = \frac{\vdash p}{\neg p} \right\}.$$

Put $E_2 = Th(\{p\})$.

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- $\Gamma(E_2) = Th(\{\dots\})$.
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Hence $\Gamma(E_4) = Th(\emptyset) \neq E_4 - E_4$ is not an extension of (A, Δ) .

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Hence $\Gamma(E_4) = Th(\emptyset) \neq E_4 - E_4$ is not an extension of (A, Δ) .

There is no other candidates for an extension of (A, Δ) .

(A, Δ) has NO extension !

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- (A, Δ) has an inconsistent extension iff A is inconsistent.
- If (A, Δ) has an inconsistent extension, this is its only extension.
- If E and F are extensions of (A, Δ) and $E \subseteq F$, then $E = F$ (**maximality of extensions**).

Normal defaults

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- **Semi-monotonicity:** Let $T_1 = (A, \Delta_1)$ and $T_2 = (A, \Delta_2)$ be closed normal default theories such that $\Delta_1 \subseteq \Delta_2$ and let E_1 be the extension of T_1 . Then T_2 has an extension E_2 such that $E_1 \subseteq E_2$.

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- **Orthogonality of extensions:** Let E and F be two different extensions of a closed normal default theory T . Then $E \cup F$ is inconsistent.

Thank you for your attention!

Any questions are welcome.